



MSc Project Report
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**Machine Learning Prediction of Intensive
Care Length of Stay for Lung Cancer
Patients**

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Abstract

Objective: This study aims to train prediction models to predict the length of stay for lung cancer patients in the **Intensive Care Unit (ICU)**. Two models will be trained, evaluated and compared for their performance and the variable importance.

Methods: Data was sourced from the **Medical Information Mart for Intensive Care (MIMIC)-IV** dataset which consists of hospital data, including emergency department data, from the Beth Israel Deaconess Medical Center in Boston, Massachusetts. Two regression models were developed, gamma **Generalised Linear Model (GLM)** and random forest. The models were evaluated by the **Root Mean Squared Error (RMSE)**, the **Mean Absolute Error (MAE)** and the R^2 statistics, via 5-fold cross validation using **Multivariate Imputation by Chained Equations (MICE)** to impute the missing values in the data. The variable importance was also investigated for the two models.

Results: 2115 patient records in the **MIMIC-IV** database met the eligibility criteria and were included in the study. Out of the 12 variables with missing data in the dataset, 4 were omitted due to exceedingly high proportions of missingness. The mechanism of missingness was investigated for the remaining 8 variables each of which showed plausibility of **Missing at Random (MAR)** and excluded the possibility of **Missing Completely at Random (MCAR)**. Both models output very similar performance values for the **RMSE**, **MAE** and R^2 after 5-fold cross validation. The variable importance graphs showed differing results for the two models, although both showed the calcium gluconate variable to be the most important.

Conclusion: Both models demonstrated poor performance and neither of the models demonstrated a clear predictive advantage over one another. Neither models would realistically be reliable for real world ward management, but if one were to be recommended it would be the gamma **GLM** as it is the more interpretable of the two. There were several limitations that reduced the models' predictive performance. In regards to missingness, a significant amount of missing data obstructed the models from providing robust predictions. The plausibility of **Missing Not at Random (MNAR)** could not be ruled out, therefore the quality of the imputations was unknown. Further limitations were observed, addressing these in future study may produce models with a higher level of predictive performance.

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List of Abbreviations

ADASYN Adaptive Synthetic Sampling Approach for Imbalanced Learning. 33

AUC Area Under the Curve. 4, 5, 6

AUPRC Area Under the Precision-Recall Curve. 5

AUROC Area Under the Receiver Operating Characteristic. 5

CART Classification and Regression Decision Tree. 17

CITI Collaborative Institutional Training Initiative. 3

CSV Comma-Separated Values. 10

ED Emergency Department. 3, 6

g/dL Grams per Decilitre. 11, 27

GLM Generalised Linear Model. i, iii, iv, v, 4, 7, 14, 15, 16, 22, 29, 30, 31, 32, 33, 35

ICD International Classification of Diseases. vi, 9, 10, 24

ICU Intensive Care Unit. i, iv, 1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 14, 24, 25, 32, 33, 34, 35

INR International Normalised Ratio. 8, 28

ku/L Kilounits per Litre. 11, 26

LOS Length of Stay. v, 1, 2, 3, 4, 6, 7, 8, 9, 14, 16, 17, 20, 22, 23, 27, 29, 31, 32, 33, 34, 35

MAE Mean Absolute Error. i, 19, 20, 29, 32

MAPE Mean Absolute Percentage Error. 4

MAR Missing at Random. i, 13, 14, 28, 34

MCAR Missing Completely at Random. i, 13, 14, 28

mEq/L Milliequivalents per Litre. 11, 27

mg/dL Milligrams per Decilitre. 11, 26

MICE Multivariate Imputation by Chained Equations. i, 14, 21, 28, 34

MIMIC Medical Information Mart for Intensive Care. i, ii, iii, 3, 4, 6, 7, 8, 9, 10, 12, 13, 24, 33, 34

MNAR Missing Not at Random. i, 13, 14, 28, 34

MSE Mean Squared Error. 17, 23

ng/L Nanograms per Litre. 11

NICE National Institute for Health and Care Excellence. 4

OLS Ordinary Least Squares. 4, 15, 16, 33

PDMS Patient Data Management System. 5

PH Proportional Hazards. 4

PTT Partial Thromboplastin Time. 8, 11, 28

RMSE Root Mean Squared Error. i, 5, 19, 20, 29, 32, 33

RMSPE Root Mean Squared Percentage Error. 4

SD Standard Deviation. 26, 27

SMOTE Synthetic Minority Over-sampling Technique. 33

SQL Structured Query Language. 8, 10, 12

US United States. 1

XGBoost eXtreme Gradient Boosting. 4, 5, 6, 7, 35

1 Introduction

1.1 Background to Lung Cancer and ICU Treatment

Cancer is a disease that is defined by the uncontrollable multiplication of malignant cells, in some cases forming tumours in organs or tissues. In the **United States (US)**, and many other countries around the world, cancer has become the leading cause of death, and in the **US** specifically it has been reported that approximately 39% people will be diagnosed with any cancer in their lifetimes[1]. In 2025 it was reported that approximately 12% of cancer cases were lung cancer cases[2]. These statistics indicate that the volume of patients requiring **ICU** treatment may be significantly large and therefore the optimisation of **LOS** analysis is key for adequate resource management and patient care planning. Lung cancer in particular describes the cancerous cell growth in the lungs and is regarded as a serious health issue for both the patient and healthcare systems worldwide.

As of Joohyn Park., et al's 2019 report[3], the mean annual health care expenditure for a lung cancer patient in the **US** is \$35,141. This was significantly higher than the 4 most common cancer types that were included and approximately 8-9 times as high for a cancer free patient, \$4,484. These numbers point towards the extent of how significant the burden of lung cancer is on the **US** healthcare system. They also report that lung cancer tends to be diagnosed at more advanced stages so the survival duration is short. A separate report notes the estimated 5 year survival rate at 27% as of 2025 [2]. With a higher number of cases diagnosed at advanced stages, **ICUs** are expected to be required to administer treatment to an increasing number of patients[4].

The **LOS** in an **ICU** ward is an important piece of information for clinicians to know and if a machine learning model could predict it accurately the model would be highly useful in improving the efficiency in the intensive care process. At the time of writing, although some machine learning predictions of **ICU LOS** are being researched[5][6][7][8], the majority of healthcare systems go by clinical judgement alone[9]. One of the main reasons behind this stems from the lack of understanding regarding how to employ predictive machine learning models to effectively assist a clinician in key decision making.[10] The lack of use of these methods in practice provides a rationale for conducting this type of study.

1.2 Machine Learning

1.2.1 What is Machine Learning?

The advancement of computational power and access to large amounts of data has created an opening for the application of statistical techniques to develop machine learning algorithms to produce accurate and useful results. Machine learning works similarly to the human brain in that it takes input information and uses it to build a form of understanding of abstract concepts. Machine learning algorithms can learn from their input data to identify patterns for use in making predictions as an aid for real life decision making.

A machine learning model relies heavily on the amount of input data, the sample size, and also the quality of said data[11]. A larger sample size allows for a broader representation of trends in the data, while data quality determines the reliability and usefulness of the results.[12]

There are many aspects of the input data that determine its quality:

- The completeness or, in other words, magnitude of missingness.
- The number of predictor (independent) variables and whether they are relevant for the outcome being predicted.
- The degree to which the sample data is representative of and can be generalised to the wider population. A model trained on a representative sample can be applied more effectively on the wider population.
- Recency of data collection. Up to date records are more informative.
- Uniformity of data units.

1.3 Intended Use

Due to factors such as an aging population, medical advancements and staffing shortages, ICUs are now greater utilised for severe medical conditions such as the complications that arise from diseases like lung cancer[13]. This raises a few challenges that a robust LOS prediction could help to solve.

It is imperative for hospital staff to have an understanding of bed availability so that they can manage the influx of new patients effectively. Knowing which beds may shortly become available

helps ward managers plan ahead for incoming patients and provides an estimate of how many beds an ICU ward may need.

Accurate LOS predictions could contribute to more informed forecasts for ward's operating cost. This information could support the development of more realistic budgets and resource allocation, which could potentially reduce the likelihood of excessive spending or underfunding.

At current clinicians do not use machine learning techniques to predict LOS for lung cancer ICU patients. Their predictions or estimates primarily come down to clinical judgement. Therefore estimates may vary depending on a clinician's experience and familiarity to the patient. This prediction is also not an immediate priority for the clinician, they may have the most granular knowledge but they are more focused on administering the best possible treatment to their patients. Therefore a clinically significant machine learning prediction model could help as an aide for hospital staff to make applicable approximations of the LOS for an individual patient[14].

1.4 The MIMIC Database

MIMIC is a large relational database that consists of deidentified data collated from thousands of admissions to the intensive care unit of the Beth Israel Deaconess Medical Centre in Boston, Massachusetts[15].

The most recent version of the MIMIC database is MIMIC-IV. It includes data collected from January 2008 to December 2022, which is an extension on the previous version, MIMIC-III, which recorded data up to December 2012. Using the updated version should allow for greater relevance and accuracy of the prediction model.

The database is split into four modules, the core data, the hospital data, the ICU data and the Emergency Department (ED) data. For my study I have not included the data from the ED module to prevent data leakage from timestamp discrepancies.

Within the core module there are three tables to store the non-clinical data that can be used for linkage purposes; these are labeled admissions, patients and transfers. The hospital module is the largest, containing nineteen tables of administrative, medication and laboratory records. The ICU module includes continuous measurements and details of bedside interventions, however, this study only uses this module to determine the patient's LOS.

The MIMIC-IV data was accessed after completing a short ethics course set up by the Collaborative Institutional Training Initiative (CITI) online training programme[16][17].

1.5 Literature Review

This literature review aimed to find relevant peer-reviewed publications to compare with the findings of this study, and to inform on the existing research and the reported results found in this area. The initial search attempted to find lung cancer **ICU LOS** machine learning prediction studies focusing on a population of adults over 18 that stated a result of predictive performance. A search with these parameters in PubMed found 5 results of which none were relevant. So the search parameters were broadened. The age restriction was dropped and so was the specificity of lung cancer, as in any **ICU LOS** machine learning prediction study was considered. This search in PubMed produced 697 results of which 13 fit this criteria.

Additionally, 1 publication was found through google scholar that fit all of the initial criteria.

Author	Year	Prediction Model	Best Model	Data Source
Ilona W.M.Verburg., et al [18]	2014	Ordinary Least Squares (OLS) regression, OLS regression truncated at 30 days, OLS regression log, GLM Gaussian, GLM poisson, GLM negative binomial, GLM Gamma, Cox Proportional Hazards (PH) regression	OLS regression truncated at 30 days: R^2 - 0.208 (0.2001 to 0.215), Root Mean Squared Percentage Error (RMSPE) - 5.150 (5.061 to 5.239), Mean Absolute Percentage Error (MAPE) - 3.099 (3.053 to 3.144), Bias - 0.015 (-0.044 to 0.074)	National Institute for Health and Care Excellence (NICE)
Qiuying Chen, et al [19]	2021	Naive Bayes, Linear Regression, Decision Tree, Random Forest, Gradient Boosting Decision Tree	Random Forest: Area Under the Curve (AUC) - 0.991 (0.978 to 1.000) and 0.837 (0.766 to 0.908) in the training and validation datasets respectively	Guangdong Provincial Cardiovascular Institute
Khalid Alghatani., et al [20]	2021	Linear Regression, Linear Discriminant Analysis, Random Forest, k-Nearest Neighbour, Support Vector Machine, eXtreme Gradient Boosting (XGBoost), Multivariate Linear Regression, Support Vector Regression	Random Forest	MIMIC-III

Author	Year	Prediction Model	Best Model	Data Source
Longxiang Su., et al [21]	2021	Logistic Regression, Random Forest, XGBoost	Random Forest: AUC - 0.76	ICU unit of Peking Union Medical College Hospital
Andrea Stieger., et al [22]	2025	Logistic Regression, Random Forest, Stacked Learner, Gradient Boosting Machine, Neural Networks	Stacked Learner: Area Under the Receiver Operating Characteristic (AUROC) - 0.94, Area Under the Precision-Recall Curve (AUPRC) - 0.62	Seoul National University Hospital
Louise Y.Sun., et al [23]	2020	Logistic Regression	Short stay: c-statistic (0.78/ 0.71, derivation/validation), Long stay: c-statistic (0.85/0.78, derivation/validation) This essentially concludes the idea that the logistic regression model may help practitioners	University of Ottawa Heart Institute
Kan Wang., et al [24]	2022	Linear Regression, Naive Bayes, Random Forest, Support Vector Machine, XGBoost	XGBoost: AUC - 0.88	Wuhan Union Hospital
Pepler PT., et al [25]	2012	Poisson Log-linear, Regression Model	Logistic Regression - AUC 0.8507	Vermont Oxford Network
Yucai Hong., et al [26]	2020	Multivariate Regression Model	Multivariate Regression Model: AUC - 0.848	Tertiary Care Hospital in Yueqing City, Zhejiang Province
Geert Meyfroidt., et al [27]	2011	Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, Gaussian Forest	Gaussian Processes: RMSE - 0.408, AUC - 0.758	MetaVision Patient Data Management System (PDMS) Database

Author	Year	Prediction Model	Best Model	Data Source
Waqar Aziz., et al [28]	2024	Random Forest, Gradient Boosting	Gradient Boosting: AUC - 0.7243	MIMIC-IV-ED
Belal Alsingwali., et al [29]	2022	Random Forest, XGBoost , Logistic Regression	Random Forest: k-fold = 10 mean accuracy - 87.4%	MIMIC-III
Ha Na Cho., et al [30]	2025	Linear Regression, Random Forest, Support Vector Machine, XGBoost , Their own developed model called SurgeryLSTM	SurgeryLSTM: R^2 - 0.86 followed by XGBoost - R^2 - 0.85	University of California Irvine Health
Taghi Khanijev., et al [31]	2025	Logistic Regression, Decision Tree, Random Forest, Neural Networks	Random Forest: AUC - 0.824	MIMIC-IV

Table 1: Previous work concerning building machine learning models to predict a patient's **ICU LOS**

From inspection of the table, it seems that the majority of these previous studies focus on building a model to predict via classification. In contrast, this study will define **LOS** as a continuous variable rather than a categorical one in which case a regression approach will be adopted. Splitting the variable into categories can make results more interpretable and more simple to visualise but doing so comes at the cost of losing precise information[32]. which may result in a loss of clinical usefulness of the predictions to make informed estimations on bed allocation and staffing needs.

Most of the studies in the table compare multiple machine learning models. The model that appears as the strongest most frequently is random forest with **XGBoost**/gradient boost following close behind. These methods are therefore worth considering as a predictive application within the lung cancer **LOS** context.

Another observation from the literature is the frequent use of the **MIMIC** database in the absence of locally sourced data. Which shows that **MIMIC** is certainly a viable source of data. Additionally many older studies use **MIMIC-III**, therefore, this study can benefit from the more recent **MIMIC-IV**.

The majority of these studies focus on building models for LOS prediction in various health related contexts. There is only one study found in the literature review that particularly focuses on lung cancer ICU LOS prediction[29]. This study used MIMIC-III data to compare random forest, XGBoost and logistic regression. The selected variables from their study were adopted as candidate variables for this study, providing consistency with similar previous research while assessing these variables on the updated version of MIMIC.

Since there is a scarcity of studies in this area, there is great benefit in further research to attempt to find a reliable and effective predictive model suitable for use in clinical settings.

1.6 Aims and Objectives

1.6.1 Aim

The main aim of this project is to build accurate and efficient machine learning models using data from the MIMIC-IV dataset to enhance the prediction of the LOS in the ICU, for lung cancer patients. The ideal result would be for these models to assist in the management of lung cancer patients in intensive care wards in terms of cost and effectiveness.

1.6.2 Objectives

The main objectives are the following:

- Extract the data from the MIMIC-IV database to form a final dataset ready to be used to develop predictive models.
- Train the random forest and gamma GLM predictive models via 5-fold cross validation. Missing data imputation will occur within the cross validation process.
- Evaluate the values quantifying each model's performance and variable importance and compare them against each other.

2 Methods and Materials

2.1 Variable Selection

Variables to use in the prediction models were determined before any data extraction from the **MIMIC-IV** database occurred. The importance of choosing relevant variables is significant as this has a strong impact on the performance of the model. Therefore some prior knowledge on the key variables that affect lung cancer is beneficial, to obtain this it would be useful to draw on medical knowledge. In the absence of first hand medical knowledge some research on the variables used in previous studies were examined. A study with the same aim of prediction for the **LOS** for lung cancer patients using the **MIMIC-III** database was identified [29]. Their variable selection process involved first considering clinical significance, followed by a recursive feature elimination to remove less informative variables. The variables that remained are as listed: age, temperature, platelets, admission type, glucose, chloride, potassium, pulse oximetry, **Partial Thromboplastin Time (PTT)**, anion gap, respiratory rate, haemoglobin, troponin, calcium gluconate, haematocrit, tidal volume, diastolic, ondansetron, **International Normalised Ratio (INR)**, sodium. Of these pulse oximetry, anion gap, respiratory rate, tidal volume and diastolic were not collected due to lack of data and haematocrit was removed due to its significant association with the haemoglobin variable. The variables sex, ethnicity and health insurance were also included based on their potential clinical significance.

The target variable, **LOS**, is to be treated as a continuous variable measured in days.

2.2 Database Management

PhysioNet[33] is a publishing platform that holds, among other things, de-identified biomedical and clinical data. pgAdmin 4 is a graphical user interface that can be used to interact with PostgreSQL, an open-source database system[34].

In order for data to be extracted from the **MIMIC** database, the data files from PhysioNet should be downloaded and exported into a database schema in pgAdmin 4. From there the database can be queried with **Structured Query Language (SQL)**[35] commands for data extraction.

To create the schema, i.e. the layout and structure of the database being created, an **SQL** script is run, specifically tailored for **MIMIC-IV**, to construct empty tables ready for the data export process. The script was retrieved from a Github repository.[36]

After creating the empty database, the data files are prepared ready for export. Some data files were too large for pgAdmin to export in one process, therefore the larger data files were split into smaller parts with the use of the Windows powershell. Once larger data files had been split they were exported into their respective tables in the schema.

2.3 Eligibility Criteria

To be included in the study patients must meet the following eligibility criteria.

- The patient must be an adult, over 18, at the time of their admission into hospital.
- The patient must have at least one **ICU** admission within their hospital stay.
- The patient must have a relevant **ICD-9** or **ICD-10** code that indicates a diagnosis of lung cancer.
- The patients' stay must be the first in their admission.

2.4 Sample Size

To calculate the sufficient minimum sample size for a prediction model development with a continuous outcome the R[37] package "pmsampsize" was used. The calculation was based on the mean and standard deviation of the outcome variable, a priori R^2 value obtained from previous study and the number of candidate predictors, where a categorical variable has multiple candidates depending on the number of categories. The priori R^2 value used for this calculation will be the R^2 obtained in Ilona W.M. Vernberg et al's study focused on developing regression prediction methods for predicting **ICU LOS**[18]. The priori R^2 is valued at 0.2.

2.5 Data Extraction

Now that the **MIMIC** database has been setup the data for the determined variables can be extracted.

To obtain the cohort from **MIMIC** for analysis, the **ICD-10** [38] codes corresponding to the diagnosis of lung cancer are needed to identify the patients required for the study. **ICD-10** codes represent various patient diagnoses in alphanumeric form, making it easier to record diagnostic data, thereby reducing the likelihood of incorrect entries. The **ICD-10** codes that indicate a patient

has lung cancer are the C34 codes, identifying a "Malignant neoplasm of bronchus and lung", or the C78.0 codes, identifying a "Secondary malignant neoplasm of lung". A secondary malignant neoplasm is a cancer that develops as a result of a previous cancer. Along with the ICD-10, MIMIC-IV also uses ICD-9 to identify health conditions. The ICD-9 codes that correspond to the lung cancer cases are the 162.9 codes. ICD-10 codelist is the updated version of the ICD-9. ICD-10 was introduced in 2015 and MIMIC-IV was collected between 2008 and 2018, which is why both sets of codes appear.

ICD Coding System	ICD Code	Condition Description
ICD-9	162.X	Malignant neoplasms of the trachea, bronchus and lung. I.e., cancer that originates in the lung
ICD-10	C34.X	Malignant neoplasms of the bronchus and lung
ICD-10	C78.0	Secondary malignant neoplasm of lung. I.e., lung cancer that develops as a result of a previous cancer

Table 2: ICD codes to identify a lung cancer diagnosis

Through pgAdmin 4, SQL queries were written to sift through the database to extract the desired data to form the final dataset.

Similarly, to find data for each variable, SQL scripts were developed to create datasets for all eligible patients in the database. The resulting datasets were saved into Comma-Separated Values (CSV) files and imported into a single R file where they were merged using a unique patient ICU identifier.

2.6 Covariates

To predict length of stay the model will need relevant data as its input. Data from variables of interest are collected, these are referred to as the covariates. The table below gives the list of covariates to extract from MIMIC, selected for their use in relevant literature identified in the literature review section.

Variable Name	Unit of Measure	Data Type	Description
Age	years	Continuous	The number of full years the patient has lived
Temperature	°C	Continuous	The closest temperature recording before admission into the ICU
Platelets	Kilounits per Litre (ku/L)	Continuous	Cell fragments that allow the blood to clot and injuries to heal
Chloride	Milliequivalents per Litre (mEq/L)	Continuous	An essential electrolyte that works in a capacity to maintain pH balance, help digestion and cell function among other uses
Sodium	mEq/L	Continuous	Sodium works together with chloride in many bodily functions. It is also a major nutrient involved in water retention and blood pressure regulation
Potassium	mEq/L	Continuous	Electrolyte that is useful to lower blood pressure by removing excess sodium. It also promotes bone density and healthy muscle function
Glucose	Milligrams per Decilitre (mg/dL)	Continuous	Responsible for energy production and brain function
Haemoglobin	Grams per Decilitre (g/dL)	Continuous	Important for the transferral of oxygen from the lungs to the rest of the body and takes carbon dioxide from cells to the lungs for it to be exhaled
PTT	seconds	Continuous	The time taken for blood to clot
Troponin	Nanograms per Litre (ng/L)	Continuous	Troponin helps the heart muscle contract and relax
Calcium Gluconate and Ondansetron	no unit	Binary	Calcium Gluconate and Ondansetron are both medication. Ondansetron aims to prevent nausea and vomiting caused by cancer treatments or surgery and Calcium Gluconate is used in cases where calcium is too low, potassium is too high and/or magnesium is too high

Variable Name	Unit of Measure	Data Type	Description
Ethnicity	no unit	Categorical	Categories of universally recognised ethnic groups: White, Black, Asian, Hispanic/Latino, Other.
Sex	no unit	Binary	Two binary categories, male and female, as recognised in MIMIC-IV
Admission Type	no unit	Categorical	Categories that represent the type of admission for the patient: Urgent, Observation, Emergency, Elective.
Health Insurance	no unit	Categorical	The type of health insurance the patient has: Medicare, Medicaid, Private, Other.

Table 3: Variables that have been extracted from **MIMIC-IV** that will be considered as predictors in the prediction models for this project

An essential data pre-processing step to ensure valid interpretations is to scale continuous data so that for each variable the mean is 0 and the standard deviation is 1. Doing so places each variable on the same scale where the coefficients correspond to the effect of a 1 standard deviation change[39]. Dummy coded categorical variables cannot be placed on this scale, however. Their coefficients represent the effect of being in the category as opposed to being in the reference category.

When multiple measurements were recorded for a continuous covariate prior to a patient's first **ICU** admission, the value closest to the time of admission was selected.

2.7 Data Cleaning

SQL and R programming were used to handle data cleaning.

Some patients had more than one **ICU** admission in their hospital stay, i.e. they were discharged and readmitted. The subsequent **ICU** stays cannot be included within the data as this would violate the assumption of independence[40] which contributes to bias in the prediction models. Including them would also cause some data leakage across the randomly split cross validation folds.

Some of the categorical variables, extracted from **MIMIC-IV**, contained many categories with some representing only a small number of individuals. For example the ethnicity variable contained several variables that contained a few individuals. To remedy this, each category was incorporated into one of 5 broad ethnic groups, White, Black, Asian, Hispanic/Latino or Other, ensuring each group contained a sufficient amount of data. The data identified as missing were labeled in the same way that missing values in the continuous variables were.

2.8 Missing Values

Missing values in the data act as a roadblock to meaningful analysis. Many machine learning models require the input data to be complete for them to work, and their performance is greatly improved when the missingness is remedied so that the resulting study population matches the true population distribution as closely as possible[41].

2.8.1 Missing Data Types

There are three distinct types of missing data, **MAR**, **MNAR** and; **MCAR**. Data is considered **MCAR** when the missingness of the data is random[42]. This differs from **MNAR** which describes missing data that depends on the context behind the missing value. For example people are less likely to disclose their substance use due to societal judgement and/or its legality. And finally **MAR** data is dependent on data from other variables[43]. The importance of these distinctions lie in the determination of the choice of imputation or deletion method of dealing with this missingness.

Dealing with missing data can involve the complete case analysis, omission of all missing data[44], or imputation. Although complete case analysis is the simpler method, it may introduce bias into the resulting model if the missing mechanism of the missing data is not **MCAR**[45]. Therefore, omission of these data could potentially lead to a sample that is misrepresentative of the population which in turn leads to the aforementioned bias in the prediction models[46].

Imputation can provide a more appropriate approach for reducing potential bias introduced by omitting missing data. Imputation is, in simplistic terms, the replacement of missing data with estimated values to provide a full dataset for analysis. The imputed data values are estimated using the information from the observed data, such as the variables' distributions and relationships with each other[47].

2.8.2 Imputation Methods

Single imputation methods were considered inappropriate in the context of this study as they impute the missing values with a single estimate which fails to account for uncertainty in the imputation process[48]. Since there is only one imputed value, single imputation can underestimate variance and lead to biased estimates particularly when the proportion of missingness is high[49].

MICE, however, more appropriately reflects the uncertainty in the imputations and the underlying variability in the data, which allows for it to produce more statistically robust estimates. This method achieves this by using data from the other variables to iteratively impute, via predictive models, until convergence of the distribution of the imputed variables[50].

Each variable used in training the model will be used in the imputation process[51]. It is possible to include auxiliary variables, but they must meet certain criteria; there will however not be any exploration of this in this study.

Logistic regression models for each incomplete variables were fitted to determine how well the other variables explained the probability of missingness. The models used a binary indicator to represent the status of missingness. If none are found to be significant then data being **MCAR** is possible although missingness being **MAR** or **MNAR** is also plausible, there just may be unobserved variables that suggest this. If some of the variables in the model are significantly associated with the missingness then the idea that the missing data is **MAR** is more plausible, although **MNAR** data may still exist.

The **MICE** method works most efficiently under the assumption that the reason of missingness is **MAR**. There is no test that can prove whether the missing data in this study is **MAR** or **MNAR** as both are plausible within a clinical setting. The study will continue with the assumption that the missing data is **MAR** even with the knowledge that **MNAR** data may be present. This is listed as a limitation in the discussion section.

2.9 Prediction Models

The primary outcome of interest in this study is the lung cancer patients' **LOS** in the **ICU**. Therefore, the performance of two prediction models, gamma **GLM** and random forest, will be evaluated to determine their suitability to predict the **ICU LOS**.

2.9.1 Gamma GLM

Linear regression is, compared to other widely known machine learning prediction methods, somewhat simple and easy to interpret. It is often used as the first step in many attempts to apply machine learning techniques for prediction purposes. Arshad et al show how a simple linear model can be more effective than other more complex models, with the example of a study involving breast cancer diagnosis prediction[52].

OLS has assumptions that ensure that the predictions are reliable and valid. Two of these assumptions are that the outcome variable data should be normally distributed and there should be no heteroscedasticity[53]. A normal distribution is one that is described by a bell curve with its mean median and mode all at the centre. Heteroscedasticity occurs when variance of the residuals either increases or decreases as input values differ[54].

The linear regression equation is expressed below.

$$Y = X\beta + \epsilon$$

Y represents the vector of predictions, X is the matrix of input values, β are the coefficients for the model and ϵ is the error term.

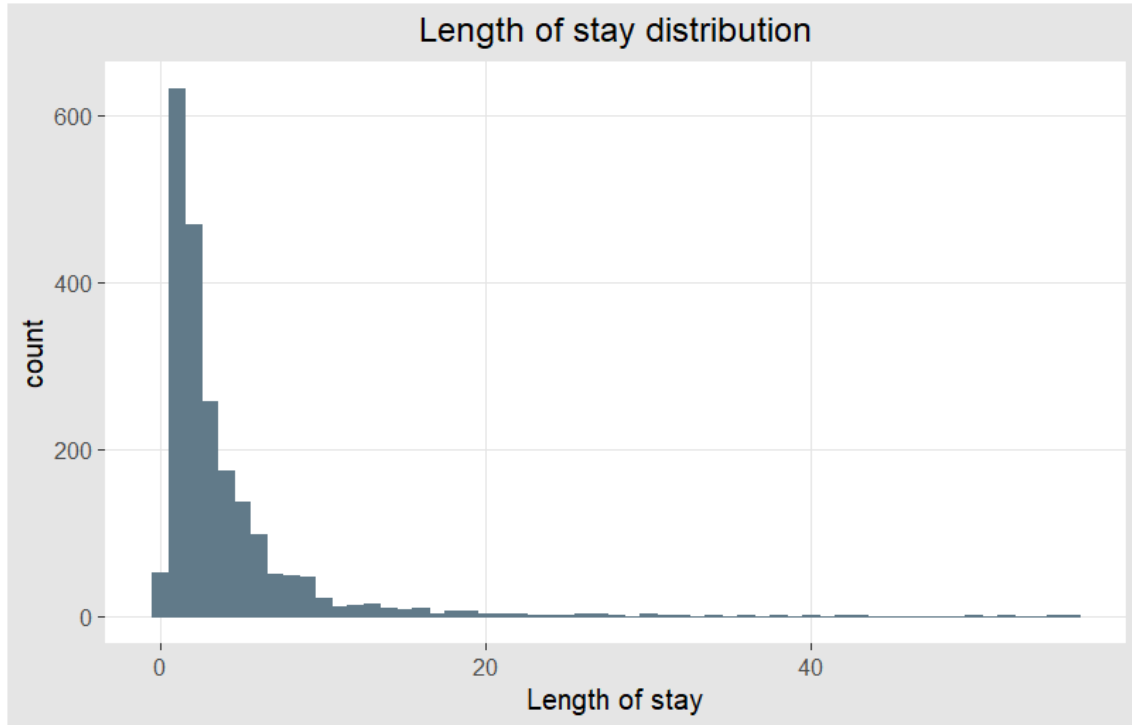


Figure 1: Shows the distribution of the los variable. Indicates that a regression model that assumes a normal distribution when used on right skewed data may not perform well. Therefore others such as the gamma glm may be a better fit.

This data's right skewed distribution violates the normality assumption rendering the **OLS** model invalid for prediction. A gamma **GLM** model is a much better fit as it is explicitly designed for right skewed data and data that is strictly positive[55], which is the case for **LOS**. Furthermore the gamma **GLM** does not assume that the model exhibits zero heteroscedasticity, so there is no need to check for this.

$$Y \sim (\mu, \phi)$$

The **LOS** data are modelled by the gamma distribution with mean μ and dispersion parameter ϕ . This accomodates an increase of variance as the expected **LOS** increases.

$$\log(\mu) = X\beta$$

This equation shows how the logarithm of the expected **LOS** μ is linked as a linear relationship which is useful for **LOS** data as it ensures that predicted values are positive.

The multiplicative effect[56] coefficient for a predictor represents a proportional change in the expected LOS. For example a coefficient of 1.02 would correspond to a 2% change in expected LOS for an increase of size 1 in the variable. Therefore, the absolute change in LOS is larger for long stays than it is for short stays.

2.9.2 Random Forest

To explain the random forest method, the concept of a **Classification and Regression Decision Tree (CART)** must be established first[57]. At the beginning of constructing a decision tree 100% of the data constitutes what is labelled the root node[58]. The root is split into two groups/child nodes, according to an attribute of one independent variable[59], so that the overall mean squared value is optimally minimised[60]. The splitting continues until a stopping point is reached. This may be because the **Mean Squared Error (MSE)** minimisation offers insignificant improvements or when the process has concluded in a predetermined maximum number of splits. As a result the data in the terminal nodes, otherwise known as leaf nodes, are relatively homogeneous with respect to the outcome.

Each terminal node includes a subset of the overall population wherein the mean for the values of the target variable constitutes their predicted value [61]. Therefore, when evaluating the model, the test data is fed into the tree and their predictions are determined according to which terminal node they fall into. The evaluation metric can then be calculated using these predictions in comparison to the actual values.

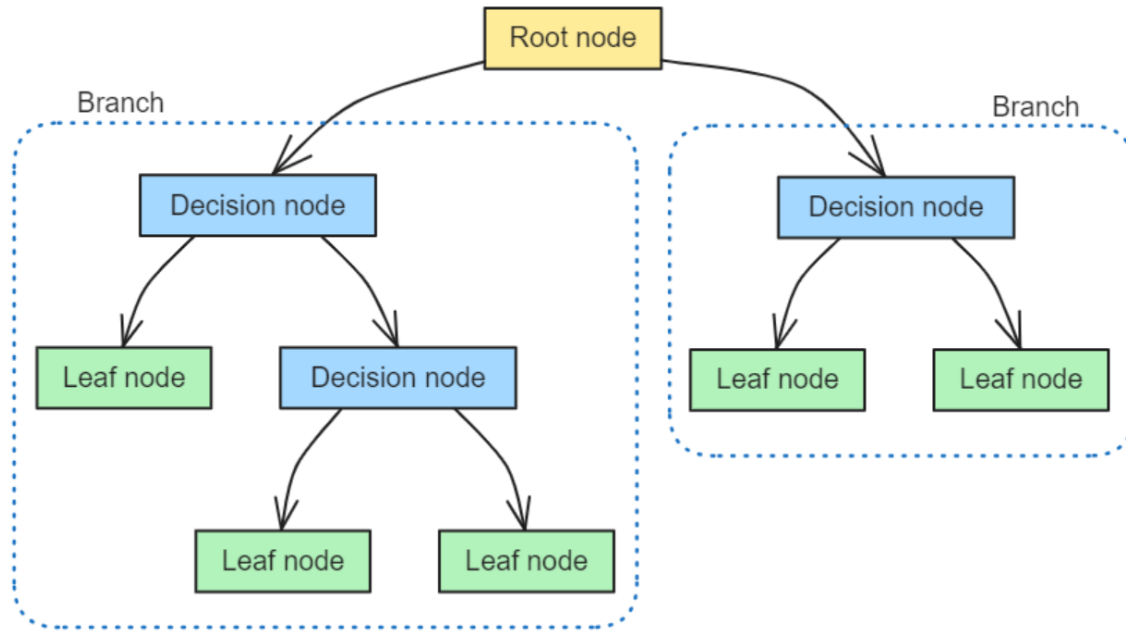


Figure 2: A diagram showing how a decision tree splits the data so that homogeneous groups form at the leaf/terminal nodes. Diagram found in figure 6 of this paper[62]

A single decision tree achieves low training error due to its high flexibility, meaning that the model is more likely to fit to noise in the data. Thereby the model is prone to overfitting. The lack of generalisability on unseen data, especially when making predictions on the test set, is a major drawback for using a single decision tree as a prediction model. To mitigate overfitting and reduce variance, multiple decision trees can be aggregated within an ensemble framework.

One such popular ensemble method is the random forest algorithm[63]. The final predictions that are obtained from a random forest model are averages from the predictions made by multiple independent, parallel decision trees[64]. The data used in each decision tree is sampled with replacement, a method known as bagging (bootstrap aggregation), to produce sets of training data each different from one another. Since each tree sees different data, their individual predictions are less correlated. Averaging the predictions across the trees reduces the variation in the model and therefore makes it less sensitive to random variation in the training data. Another feature of the algorithm is random variable selection. At each split only a subset of variables are considered. This promotes diversity among the trees by reducing the correlation between trees[65], contributing to a more robust model.

2.10 Evaluation of Prediction Models

It is important to consider the validity of prediction models. There are several methods that are designed to check the generalisability of how well models fit to new data which can be assessed using internal and/or external validation methods[66]. A prediction model requires appropriate validation so that it can be assessed for suitability in practical use. Validation also allows the predictive performance of different models to be compared.

2.10.1 Internal Validation

A common method of internal validation is k-fold cross-validation[67]. This involves splitting the dataset into k folds, usually k is set to 5 or 10, one split is used as the test dataset and the others used to train the model. In this way, all data is used both to train and test. The performance metrics in each fold can be evaluated using the predictions and actual values for the test set. 5-fold cross-validation was selected to repeatedly assess the model performance therefore reducing the variability that is associated with a single fold cross-validation[68].

Other validation methods are available to use but they come with significant downsides. For example the jackknife and bootstrap methods are not suitable because they are much less effective for larger datasets due to their high computational complexity and computational cost[69][70]. The 5-fold cross-validation's computational intensity is relatively manageable and it works well with larger datasets.

The best way to validate a machine learning prediction model is to validate externally. A model validated only on the data included in the original dataset still carries a lot of optimism and may fit poorly. Evaluation on relevant data obtained in new settings gives a better estimate of how generalisable the model is to a diverse range of populations[71]. Due to unavailable external data external validation will not be explored in this study.

2.10.2 Performance Measures

The **RMSE** is a metric that measures the standard deviation of the residuals in the model. An advantage of using this metric is that it is in the same units as the target variable adding ease to interpretability. In the context of these models the **RMSE** represents the typical magnitude of the prediction error in days[72].

MAE shows a similar measure as **RMSE** in that it measures how much a model's predictions

deviate from the actual values. The difference between the two is how this figure is calculated as **MAE** is the absolute average error value.

RMSE is more sensitive to larger errors because the errors are squared before they are averaged. Thereby the **RMSE** metric is more useful when assessing the risk of long **LOS**, which is important because long stays can be very costly. However the **MAE** is more useful for the aim of estimating the average error for the typical patient because long stays will not so heavily affect the prediction error.

The metric R^2 measures the proportion of the variance that can be explained in the outcome variable by the predictor variables which helps determine the model's predictive usefulness alongside the other performance values[73].

2.10.3 Cross Validation to Imputation Pipeline

To evaluate the performance of the models it's important to consider the order of pre-processing (imputation) and cross validation. The order is important for reducing data leakage[74], which introduces optimistically biased predictions of the observed data causing the reduction of the suitability for use of the model to make predictions on unseen data.

By imputing first, the data is imputed only once, so any data used in validation affects the way in which the missing data is imputed thereby leading to a significant amount of data leakage. In comparison by using an imputation within each fold, the data leakage is reduced because the data from the test dataset never affects the imputation of the data used to train the model. So in terms of avoiding data leakage the order of imputation within the cross validation is preferable; the evaluation that imputation within cross validation outputs is less optimistically biased and closer to the true evaluation to the model's ability to predict accurately on unseen data.

So with the decision made that the order of imputation within cross validation as the more suitable choice a visualisation is useful to properly show how this works.

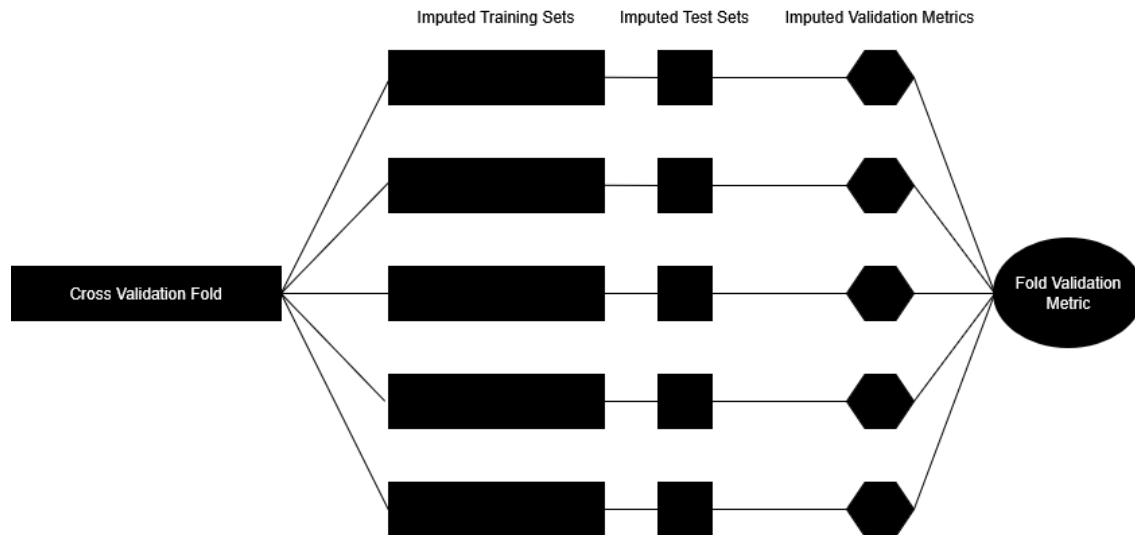


Figure 3: This shows how, within one fold of the outer cross validation, the dataset is imputed producing five independent datasets. By Rubin's rules each dataset is split into a training and test set, the model is trained and evaluated in each imputed variant of the data. The validation performances are then pooled together to output the validation performance for the fold.

This visualisation shows how to implement the **MICE** imputation process within one cross validation fold to produce a value of the validation metric. The cross validation fold with missing data is imputed five separate times. Within each fold, the train set is imputed first, then the corresponding test set is imputed using the imputation model fitted on the training set. Each test-train pairs are then validated as shown in the figure below. Each imputed dataset produces a validation metric, and these metrics are averaged to obtain a single validation value for each fold. The process of creating several different imputed datasets and averaging the performance metrics follows Rubin's rules[47] which preserves the uncertainty of missing data imputation.

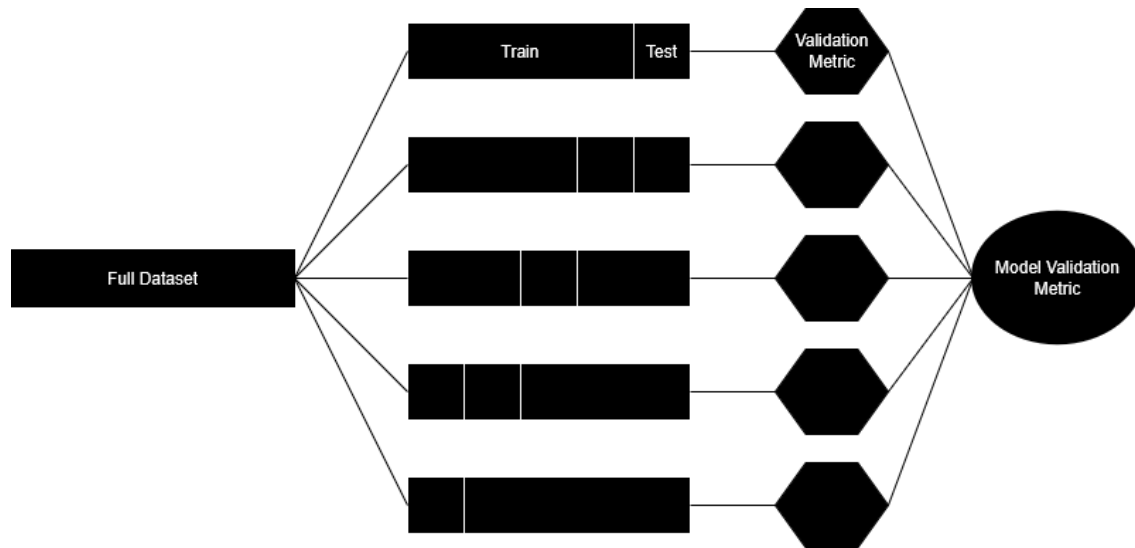


Figure 4: 5-fold cross validation where the validation performance value for each fold is calculated using the method in figure 3

The incomplete dataset is partitioned into five folds. In each iteration, one fold is held out as the test set, while the remaining four folds are used as the training set. Within each fold, the full modelling pipeline (as described in the previous figure) is carried out, including any preprocessing, imputation, and model fitting procedures. This produces one performance value per fold. The five fold-level validation metrics are then averaged to obtain the final estimate of model performance.

2.11 Variable Importance

2.11.1 Gamma GLM

The forest plot will show each variables multiplicative effect on the LOS prediction. Therefore a value of 1 would suggest that a variable's increase or decrease would not affect the value of the prediction. This leads to the fact that a variable would be more impactful/important the further away its multiplicative coefficient is away from 1, either lower or higher. For example, a multiplicative coefficient of 1.2 indicates that, when holding all other predictors constant, an increase of 1 in this variable is associated with a 20% increase in the expected LOS. Alternatively a multiplicative coefficient of 0.8 is associated with a 20% decrease.

The forest plot is primarily used to interpret individual predictor effects rather than rank variables by predictive performance, although the effect estimates may be compared. With these values the GLM model provides clear interpretability[75] for each variable's estimated effect on the LOS

outcome as there is a distinct figure that shows the magnitude and direction of the association.

2.11.2 Random Forest

Within each random forest model bootstrap samples of the inner training fold datasets are used to train each individual tree. The data not included in the bootstrap sample are termed out of the bag observations and these are used to calculate the variable importance.

To calculate permutation variable performance for the random forest model, the baseline **MSE** is first calculated for each tree using its predictions. A variable's importance is assessed by randomly permuting its values so that the relationship between the variable and the **LOS** outcome is disrupted; the other variable's values are not changed. The **MSE** is recalculated so that the difference between this **MSE** and the baseline can be obtained. This is repeated across all the trees with the average of the **MSE** taken as the measure of variable importance. A larger overall value suggests that the model relied more on this variable for an accurate prediction[76].

3 Results

3.1 Dataset Description

3.1.1 Data Extraction

MIMIC-IV provides data on 364,627 unique individuals who in total record 546,028 hospitalisation admissions with 94,458 unique ICU stays. To identify the lung cancer ICU stays the data in the ICU stays table must be sifted through by the lung cancer diagnosis ICD-9 and ICD-10 codes. Data is removed if it is outside of the normal range in any of the variables. After doing this the number of ICU stays for patients with lung cancer comes to a value of 2,390, sourced from 2,115 unique admissions. Only the first ICU stay is taken so the final dataset consists of 2,115 independent ICU stays.

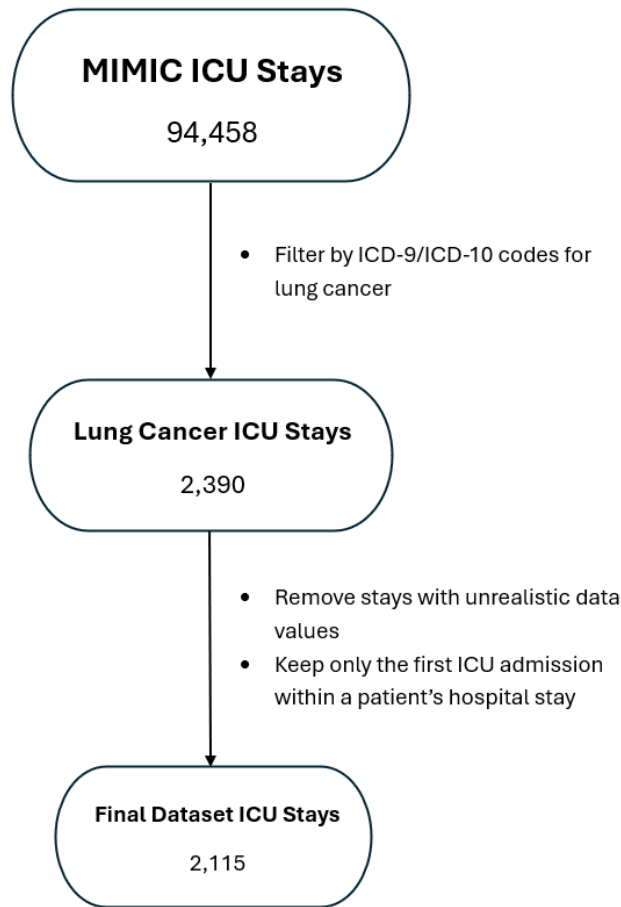


Figure 5: A flowchart showing how the sample size is obtained from the database according to the eligibility criteria

3.1.2 ICU Stay Demographics

Among the 2,115 ICU stays there is a slight favour of males, 54%, to females, 46%. The most populous age range is the 61-70 year old group, 33.4%, followed by the 71-80 group, 29.0%. The 18-40 group is markedly small, 0.6%, given its wide range. The majority of patients' ICU stays, included in this dataset, are of white ethnicity, 72.6%, and the majority of patients are in hospital with medicare insurance, 61.6%. Over half of the stays, 55.5% were from the result of emergency admissions. More detailed statistics are given in table 4.

Characteristic	N = 2115
Age (years)	
18-40	13 (0.6%)
41-50	83 (3.9%)
51-60	346 (16.4%)
61-70	703 (33.2%)
71-80	624 (29.5%)
81-90	293 (13.9%)
91+	53 (2.5%)
Sex	
Male	1,125 (53.2%)
Female	990 (46.8%)
Ethnicity	
White	1,538 (72.7%)
Black	207 (9.8%)
Asian	113 (5.4%)
Hispanic/Latino	47 (2.2%)
Other	60 (2.8%)
Missing	150 (7.1%)
Insurance	
Medicare	1,305 (61.7%)
Private	492 (23.3%)
Medicaid	262 (12.4%)
Other	40 (1.9%)
Missing	16 (0.7%)
Admission Type	
Emergency	1,186 (56.1%)
Observation	330 (15.6%)
Urgent	297 (14.0%)
Elective	302 (14.3%)
Platelets (ku/L)	
Mean (Standard Deviation (SD))	262.9 (134.1)
N missing (%)	1,325 (62.6)
Glucose (mg/dL)	

Characteristic	N = 2115
Mean (SD)	137.5 (78.1)
N missing (%)	1,235 (58.4)
Chloride (mEq/L)	
Mean (SD)	100.5 (8.4)
N missing (%)	1,241 (58.7)
Potassium (mEq/L)	
Mean (SD)	4.8 (5.3)
N missing	1,227 (58.0)
Haemoglobin (g/dL)	
Mean (SD)	10.7 (2.2)
N missing (%)	1,239 (58.6)
Sodium (mEq/L)	
Mean (SD)	135.2 (13.0)
N missing (%)	1,233 (58.3)
Drugs taken	
Ondansetron	345 (16.3%)
Calcium Gluconate	841 (40.0%)
LOS (days)	
Mean (SD)	3.8 (5.2)

Table 4: Details for each variable in the dataset

3.2 Missing Data

All of the data variables shown in this graph have highly significant amounts of missing data with not one variable showing a percentage of missing data less than 50 %. This is not very conducive for the inclusion of these variables in the prediction models since when data is more than 40 % missing it can be considered as hypothesis generating and therefore will likely not provide accurate predictions[77]. However, some data must be used so for the purposes of the continuation of the study so the variables in green are kept as predictors in the model.

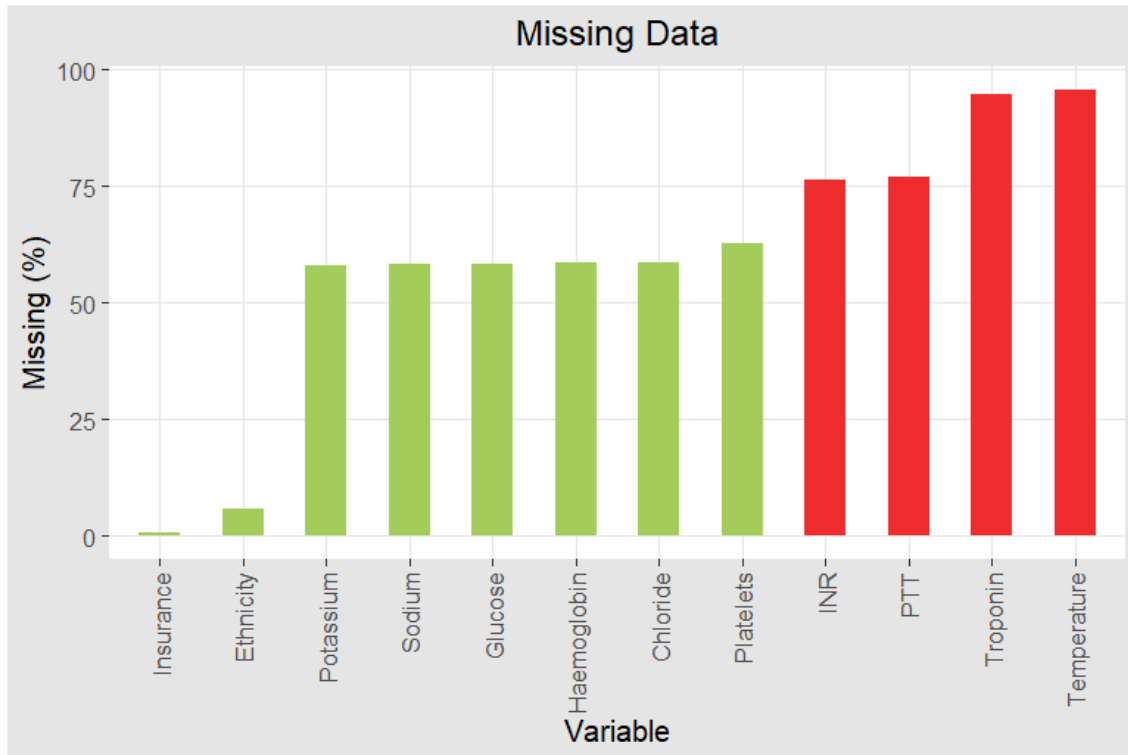


Figure 6: A bar chart showing the percentages of missing data for the numerical variables in the dataset. The variables in green will be considered as predictors in the prediction models whereas the variables in red are dismissed with INR as **INR** and PTT as **PTT**

3.3 Priori Sample Size

With the number of candidate predictors outlined above it is now possible to calculate the minimum sample size sufficient for this study with the priori R^2 assumption. 19 candidate predictors were identified by the variable selection process. Inputting the data into the "pmsampsize" R command the resulting minimum sample size is valued at 676. This is lower than the sample size of 2115 thereby suggesting that this sample is sufficient based on the assumptions made.

3.4 Imputation

The logistic regression models fitted to assess the mechanism of missingness showed that for each variable its missingness was significantly associated with at least one of the other observed variables. Therefore there is plausibility of **MAR** in each observed variable which rules out the possibility of **MCAR**. The **MICE** imputation proceeds with the assumption of **MAR** despite the inability to exclude the plausibility of **MNAR**.

3.5 Model Performance

The gamma GLM model had a narrow predictive advantage according to the MAE metric with a 0.05 day difference between its MAE of 2.86 to the random forest's 2.91. Similarly for the R^2 values both model's performance varied only slightly where the gamma GLM model explained 4.0% of the variation in the LOS variable compared to the random forest's 4.5%. Again this is true for the RMSE values, in this case the gamma GLM regression performed slightly worse with its value of 4.88 compared to the random forest's 4.86. On the whole the discrepancy between the performance values is insignificant therefore neither model appears to have a clear predictive advantage over the other.

Model	RMSE	MAE	R^2
Gamma GLM	4.88	2.86	0.040
Random Forest	4.86	2.91	0.045

Table 5: The model performance in terms of RMSE, MAE and R^2

3.6 Sample Size Sensitivity Analysis

Now that the actual R^2 performance value has been found for the two models the minimum sufficient sample size can be recalculated to assess whether the sample size was adequate for the observed model performance. The models' actual R^2 are lower than the priori R^2 value thereby doubt is raised over the sufficiency of the sample size. 0.04 was the lowest observed R^2 value so this will be used as input in the sample size calculation. The output of the "pmsampsize" calculation is valued at 3698. This minimum exceeds the sample size used in the prediction model suggesting insufficiency.

3.7 Variable Importance

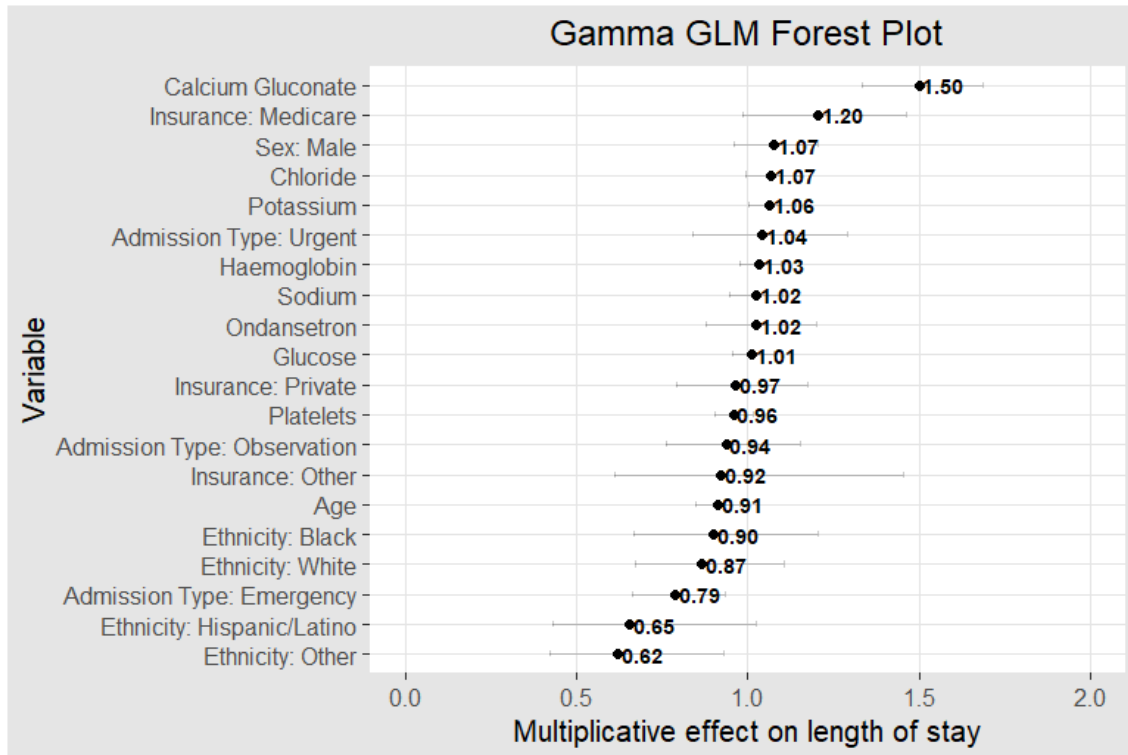


Figure 7: A forest plot showing the variable importance within the GLM model as multiplicative effect coefficients

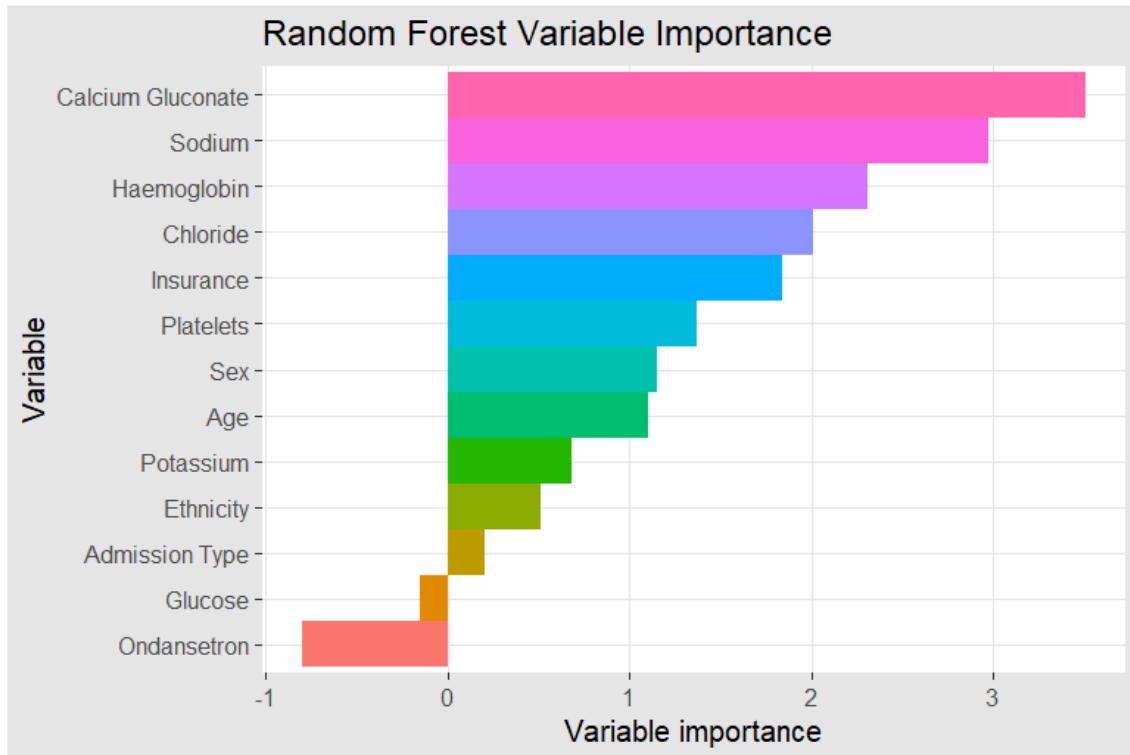


Figure 8: A chart showing the variable importance within the random forest model

Both random forest and GLM suggest that calcium gluconate as the most important variable. This suggests this variable may both have a strong association with expected LOS and a substantial contribution to predictive performance. The same comparative agreement is not preserved for the next couple of variables, however. The two variables that random forest finds to have the most predictive influence sodium and haemoglobin, which both have a relatively weaker association with expected LOS according to the GLM model. Conversely, several variables with strong LOS association in the GLM model, including the categorical variables ethnicity, admission type and insurance, demonstrated varying degrees of predictive influence in the random forest but were not among most influential predictors. These differences highlight the different ways in which the GLM and random forest evaluate variable importance, with the GLM focusing on the estimated relationships with LOS and random forest with predictive accuracy.

4 Discussion

4.1 Findings

The values from the performance metrics suggest that neither models produce robust and reliable predictions of LOS and neither model has greater predictive advantage over the other. Since the random forest model is the less interpretable[75] as there is no output equation that details the absolute effect size for each variable. Random forest can capture complex non-linear relationships and higher order interactions without needing specification beforehand,[78] but in this case that has not shown to give a distinct advantage.

The RMSE and MAE values indicated that the predicted values differed from the actual values by around 3 days on average, with larger errors contributing to an overall error of 5 days on average. The average LOS in the data is approximately 3.8 days. Given the RMSE and MAE values, the model's ability to accurately predict the LOS for short-term patients, who constitute the majority in the dataset, is limited. Therefore these models would not be suitable for supporting bed management in the ICU.

The R^2 squared values for the GLM and random forest were 0.04 indicating that the models explained 4% of the variability in the ICU LOS. This suggests that the variables included in the models provided limited explanatory power for LOS which meant that a significant amount variance remained unexplained. When considered alongside the RMSE and MAE values these findings imply that the models display a limited predictive ability for estimating lung cancer ICU LOS and so are unlikely to provide reliable predictive information that would helpfully aid clinician or bed management decisions.

The observed R^2 value being lower than the priori R^2 assumption used to calculate minimum sample size suggested that this initial calculation may have been optimistic. The sensitivity analysis found that the sample size of 2115 used in developing the prediction models was inadequate to meet the expectations of the priori R^2 performance, thereby suggesting that a larger sample size was required. Therefore it must be stated that due to this result the generalisability and reliability of the model estimates was reduced.

The calcium gluconate variable was shown in both models to be significantly important. This finding, however, is severely limited by the R^2 value of 0.04 which implied the included variables only explained 4% of variability in LOS. Therefore the definitive finding of calcium gluconate as a significant predictor of lung cancer ICU LOS is unfounded.

The substantial right skew in the LOS data poses a challenge for modelling and interpreting predictive performance. For OLS the existence of skew is often associated with a violation of the normality and heteroscedasticity assumptions.[79] The right skewed distribution of LOS is relevant to the interpretation of RMSE as it has a higher sensitivity to large prediction errors and extreme values. The decision to use gamma GLM regression to account for the skew was justified although the relatively large RMSE result suggests that this decision alone was not sufficient enough to provide accurate predictions.

4.2 Comparison to the Literature

The predictive performance for the models in this study was lower than some of the reported findings from the studies found in the literature review. However, there are many factors that could explain this difference. Firstly, there is a higher likelihood published studies to report stronger or more significant results which introduces a publication bias.[80] Many of the models in these previous studies utilised data with a lower proportion of missing data and many included a greater number of variables which may have enhanced their performance and explained a higher proportion of variance in the outcome.

The scarcity of studies using MIMIC-IV or examining patient's lung cancer ICU LOS limited the ability to draw direct comparisons with previous research. There were, however, studies using data from other databases focusing more on cancer more broadly that the findings from this study can be compared to, although the difference in the data source limits the validity of the comparisons. Many of these studies followed a classification prediction approach rather than regression used in this study. The difference in the type of performance metrics make direct comparisons of predictive performance difficult, but it does show that alternative approaches may provide more appropriate insights in regards to ICU LOS prediction.

Another takeaway from the literature is the tendency for the random forest model to display stronger and more reliable predictive performance compared to alternative machine learning methods. The trend was not shown in this study as neither the GLM model or the random forest performed better than each other.

The study that focused on lung cancer LOS prediction using the MIMIC-III using similar key predictors[29] found that the random forest model gave the most accurate predictive results. This study however focused on classification prediction and also incorporated advanced techniques such as Synthetic Minority Over-sampling Technique (SMOTE)[81] and Adaptive Synthetic Sam-

pling Approach for Imbalanced Learning (ADASYN)[82] to address the class imbalance in the data allowing the model to make robust predictions in extreme cases. The variable importance graph also showed significantly differing results, although the method used to calculate importance was not the same as the one used in this study.

4.3 Limitations

There was a substantial proportion of missing data in the dataset which can introduce several limitations including an increased risk of bias, reduced statistical power and a reduced reliability of imputation methods. As a consequence this may cause less robust and generalisable prediction models.

With clinical data there is a significant chance that the data would be **MNAR**. This is a significant limitation as the **MICE** imputation method is applicable when there is significant evidence to suggest that the data is **MAR**. With **MNAR** data **MICE** may introduce bias within the imputed data thereby rendering a potentially misleading model performance and variable importance.

The models do not account for death in the **ICU**. This confuses the model as it would treat, for example, a rapid recovery and a death in the **ICU** both with short stays as equivalent when in fact they represent two fundamentally different outcomes. A rapid recovery may suggest that the patient had a case of low severity whereas a death in the ICU may suggest the opposite. This leads to a reduced predictive accuracy and less reliable variable importance.

The models predict with a small amount of variables. This may mean that the model is too simplistic and lacking meaningful information from key indicators. Consequently the model's predictive performance would be weakened.

Although general **ICU LOS** with machine learning models is well studied, specific disease models, such as those for lung cancer, are relatively unexplored. This presents a limitation and a difficulty as a lack of prior study provides a scarcity of models to compare predictive performance and guide methodology.

MIMIC-IV is used as the sole source of data, limiting the opportunity to validate on external data in order to assess the models' generalisability across varying populations.

The sample size was shown to be insufficient for the observed model performance.

4.4 Future Recommendations

Model testing on data from new areas and alternative demographics. A model tested on one set of data from a single hospital may not generalise well to other healthcare settings where aspects such as patient demographics or medical practice may differ. External validation among various populations is necessary when assessing the robustness of the prediction model. It may be more useful to adapt the model depending on the makeup of the population that it's used on.

Work on improving data completeness before model training will allow for more reliable imputation thereby leading to improved performance and model stability.

To adjust for the event of discharge via death in the ICU two separate predictions could be made. One for if the patient survives the ICU admission and one for if they don't. Therefore the equation for LOS could be: (probability of death in ICU)(LOS given the patient dies during the stay) + (probability of survival in the ICU)(LOS given the patient survives the ICU admission).

Random forest has shown the more reliable results but further evaluation of additional machine learning algorithms may provide a more robust alternative that gives a greater model performance. Algorithms such as XGBoost and neural networks could potentially be better alternatives.

As stated previously considering new variables, such as medical complications or comorbidities, may improve model performance.

The sample size was shown to be insufficient so future research should consider sourcing a larger sample of patients. The priori R^2 value used in this study was sourced from one previous study, however, so future research should look towards finding multiple studies to provide a more informed priori R^2 value for use in estimating the minimum sufficient sample size.

5 Conclusion

Since neither model was shown to have discernable advantages in predictive performance the more interpretable model is favoured, which in this case is the GLM gamma. Findings were not clinically significant enough for either of these models to be recommended to make accurate predictions on lung cancer patients' LOS in the ICU, however, but due to the depths of the limitations within the study the possibility for an effective prediction model such as this to be created cannot be ruled out. Further study with reduced impact from its limitations may provide a more reliable prediction model.

References

- [1] *Cancer of Any Site - Cancer Stat Facts* — *seer.cancer.gov*. <https://seer.cancer.gov/statfacts/html/all.html>. [Accessed 24-04-2026].
- [2] *cancer.org*. <https://www.cancer.org/content/dam/cancer-org/research/cancer-facts-and-statistics/annual-cancer-facts-and-figures/2025/2025-cancer-facts-and-figures-acf.pdf>. [Accessed 24-04-2026].
- [3] Joohyun Park and Kevin A. Look. “Health Care Expenditure Burden of Cancer Care in the United States”. In: *Inquiry (United States)* 56 (2019). ISSN: 19457243. DOI: [10.1177/0046958019880696](https://doi.org/10.1177/0046958019880696).
- [4] Lucile Laporte et al. “Ten-year trends in intensive care admissions for respiratory infections in the elderly”. In: *Annals of Intensive Care* 8 (1 Dec. 2018). ISSN: 21105820. DOI: [10.1186/s13613-018-0430-6](https://doi.org/10.1186/s13613-018-0430-6).
- [5] Min Zhang and Tsung Ting Kuo. “Early prediction of long hospital stay for Intensive Care units readmission patients using medication information”. In: *Computers in Biology and Medicine* 174 (May 2024). ISSN: 18790534. DOI: [10.1016/j.compbiomed.2024.108451](https://doi.org/10.1016/j.compbiomed.2024.108451).
- [6] Yue Yuan et al. “Artificial intelligence-assisted machine learning models for predicting lung cancer survival”. In: *Asia-Pacific Journal of Oncology Nursing* 12 (Dec. 2025). ISSN: 23496673. DOI: [10.1016/j.apjon.2025.100680](https://doi.org/10.1016/j.apjon.2025.100680).
- [7] Ying Zhao et al. “Machine Learning-Driven Prediction of Intensive Care Units Mortality and Length of Stay: A 11-Year Retrospective Study in Hong Kong Public Hospitals”. In: *Journal of medical systems* 50 (1 Dec. 2026). ISSN: 1573689X. DOI: [10.1007/s10916-026-02355-8](https://doi.org/10.1007/s10916-026-02355-8).
- [8] Victoria Nguyen and Rahul Mittal. “Machine Learning Prediction of ICU Mortality and Length of Stay in Atrial Fibrillation: A MIMIC-IV/MIMIC-III Study”. In: *Healthcare (Switzerland)* 14 (3 Feb. 2026). ISSN: 22279032. DOI: [10.3390/healthcare14030356](https://doi.org/10.3390/healthcare14030356).
- [9] Gregory Mak et al. “Physicians’ Ability to Predict Hospital Length of Stay for Patients Admitted to the Hospital from the Emergency Department”. In: *Emergency Medicine International* 2012 (2012), pp. 1–4. ISSN: 2090-2840. DOI: [10.1155/2012/824674](https://doi.org/10.1155/2012/824674).
- [10] Marzyeh Ghassemi et al. *A Review of Challenges and Opportunities in Machine Learning for Health*. Tech. rep.

- [11] Li-Pang Chen. “Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar: Foundations of machine learning, second edition”. In: *Statistical Papers*. Vol. 60. Oct. 2019, pp. 1–2. DOI: [10.1007/s00362-019-01124-9](https://doi.org/10.1007/s00362-019-01124-9).
- [12] Steven G Johnson et al. “Quantifying the Effect of Data Quality on the Validity of an eMeasure Background and Significance”. In: (2017). DOI: [10.4338/ACI-2017-03-RA](https://doi.org/10.4338/ACI-2017-03-RA). URL: <https://doi.org/>.
- [13] Neil A. Halpern et al. “Trends in critical care beds and use among population groups and medicare and medicaid beneficiaries in the United States: 2000-2010”. In: *Critical Care Medicine* 44 (8 Aug. 2016), pp. 1490–1499. ISSN: 15300293. DOI: [10.1097/CCM.0000000000001722](https://doi.org/10.1097/CCM.0000000000001722).
- [14] Thiago Tavares Dos Santos et al. “Predicting intensive care unit length of stay: comparing physician assessments with software predictions in a multicenter study”. In: *Einstein (Sao Paulo, Brazil)* 23 (2025), eAO1265. ISSN: 23176385. DOI: [10.31744/einstein_journal/2025AO1265](https://doi.org/10.31744/einstein_journal/2025AO1265).
- [15] *MIMIC-IV v3.1* — *physionet.org*. <https://physionet.org/content/mimiciv/3.1/>. [Accessed 28-04-2026].
- [16] *CITI Course Instructions* — *physionet.org*. <https://physionet.org/about/citi-course/>. [Accessed 28-04-2026].
- [17] <https://www.facebook.com/citiprograminfo>. *Tech Ethics Training Solution* — *about.citiprogram.org*. <https://about.citiprogram.org/solutions/tech-ethics/>. [Accessed 28-04-2026].
- [18] Iwm W M Verburg et al. “Comparison of Regression Methods for Modeling Intensive Care Length of Stay”. In: *PLoS ONE* 9 (10 2014), p. 109684. DOI: [10.1371/journal.pone.0109684](https://doi.org/10.1371/journal.pone.0109684). URL: www.plosone.org.
- [19] Debasis Das et al. “Article 675431 Pradeep Narayan, Rabindranath Tagore International Institute of Cardiac Sciences (RTIICS), India (2021) Predicting Intensive Care Unit Length of Stay After Acute Type A Aortic Dissection Surgery Using Machine Learning”. In: *Front. Cardiovasc. Med* 8 (2021), p. 675431. DOI: [10.3389/fcvm.2021.675431](https://doi.org/10.3389/fcvm.2021.675431). URL: www.frontiersin.org.
- [20] Khalid Alghatani et al. “Predicting Intensive Care Unit Length of Stay and Mortality Using Patient Vital Signs: Machine Learning Model Development and Validation”. In: *JMIR Medical Informatics* 9 (5 May 2021), e21347. ISSN: 2291-9694. DOI: [10.2196/21347](https://doi.org/10.2196/21347).

- [21] Borna Relja et al. “Early Prediction of Mortality, Severity, and Length of Stay in the Intensive Care Unit of Sepsis Patients Based on Sepsis 3.0 by Machine Learning Models”. In: *Frontiers in Medicine* | www.frontiersin.org 1 (2021), p. 664966. DOI: [10.3389/fmed.2021.664966](https://doi.org/10.3389/fmed.2021.664966). URL: www.frontiersin.org.
- [22] Andrea Stieger et al. “Predicting admission to and length of stay in intensive care units after general anesthesia: Time-dependent role of pre- and intraoperative data for clinical decision-making”. In: *Journal of Clinical Anesthesia* 103 (Apr. 2025). ISSN: 18734529. DOI: [10.1016/j.jclinane.2025.111810](https://doi.org/10.1016/j.jclinane.2025.111810).
- [23] Louise Y Sun et al. “Derivation and Validation of a Clinical Model to Predict Intensive Care Unit Length of Stay After Cardiac Surgery”. In: *Journal of the American Heart Association J Am Heart Assoc* 9 (2020), p. 17847. DOI: [10.1161/JAHA.120.017847](https://doi.org/10.1161/JAHA.120.017847). URL: <https://www.ahajournal.org>.
- [24] Qingpeng Zhang et al. “Comparison of Four Machine Learning Techniques for Prediction of Intensive Care Unit Length of Stay in Heart Transplantation Patients”. In: (2022). DOI: [10.3389/fcvm.2022.863642](https://doi.org/10.3389/fcvm.2022.863642). URL: www.frontiersin.org.
- [25] FT Pepler, DW Uys, and DG Nel. “Predicting mortality and length-of-stay for neonatal admissions to private hospital neonatal intensive care units: a Southern African retrospective study”. In: *African Health Sciences* 12 (2 July 2012). ISSN: 1680-6905. DOI: [10.4314/ahs.v12i2.14](https://doi.org/10.4314/ahs.v12i2.14).
- [26] Yucai Hong et al. “Clinical characteristics of Coronavirus Disease 2019 and development of a prediction model for prolonged hospital length of stay”. In: *Annals of Translational Medicine* 8 (7 Apr. 2020), pp. 443–443. ISSN: 23055839. DOI: [10.21037/atm.2020.03.147](https://doi.org/10.21037/atm.2020.03.147).
- [27] Geert Meyfroidt et al. *Computerized prediction of intensive care unit discharge after cardiac surgery: development and validation of a Gaussian processes model*. Tech. rep. 2011. DOI: [10.1186/1472-6947-11-64](https://doi.org/10.1186/1472-6947-11-64). URL: <http://www.biomedcentral.com/1472-6947/11/64>.
- [28] Waqar Sulaiman Aziz et al. “Emergency Department Length of Stay Classification Based on Ensemble Methods and Rule Extraction”. In: *Studies in Health Technology and Informatics*. Vol. 316. IOS Press BV, Aug. 2024, pp. 1812–1816. ISBN: 9781643685335. DOI: [10.3233/SHTI240783](https://doi.org/10.3233/SHTI240783).

- [29] Belal Alsinglawi et al. “An explainable machine learning framework for lung cancer hospital length of stay prediction”. In: (2022). DOI: [10.1038/s41598-021-04608-7](https://doi.org/10.1038/s41598-021-04608-7).
- [30] Ha Na Cho et al. “SurgeryLSTM: a time-aware neural model for accurate and explainable length of stay prediction after spine surgery”. In: (2025). DOI: [10.1093/jamiaopen/ooaf079](https://doi.org/10.1093/jamiaopen/ooaf079). URL: <https://doi.org/10.1093/jamiaopen/ooaf079>.
- [31] Taghi Khaniyev et al. “Evaluating the Machine Learning Models in Predicting Intensive Care Unit Discharge for Neurosurgical Patients Undergoing Craniotomy: A Big Data Analysis”. In: *Neurocrit Care* 43 (2028), pp. 512–529. DOI: [10.1007/s12028-025-02246-9](https://doi.org/10.1007/s12028-025-02246-9). URL: <https://doi.org/10.1007/s12028-025-02246-9>.
- [32] Douglas G. Altman and Patrick Royston. *The cost of dichotomising continuous variables*. May 2006. DOI: [10.1136/bmj.332.7549.1080](https://doi.org/10.1136/bmj.332.7549.1080).
- [33] *PhysioNet*. 2026. URL: <https://physionet.org/about/>.
- [34] *pgAdmin4*. 2026. URL: <https://www.pgadmin.org/docs/pgadmin4/latest/index.html>.
- [35] *What Is Structured Query Language (SQL)?* 2026. URL: <https://www.ibm.com/think/topics/structured-query-language>.
- [36] Jennie Martin. *SQL code to load MIMIC-IV schema*. URL: <https://pages.doit.wisc.edu/JLMARTIN22/mimic-code/-/blob/sqlfluff/mimic-iv/buildmimic/postgres/create.sql>.
- [37] *R: What is R?* 2026. URL: <https://www.r-project.org/about.html>.
- [38] International Statistical Classification of Diseases and Related Health Problems. *ICD-10 Version:2019*. 2019. URL: <https://icd.who.int/browse10/2019/en>.
- [39] Andrew Gelman. “Scaling regression inputs by dividing by two standard deviations”. In: *Statistics in Medicine* 27 (15 July 2008), pp. 2865–2873. ISSN: 02776715. DOI: [10.1002/sim.3107](https://doi.org/10.1002/sim.3107).
- [40] Phillip E. Pfeifer. “Assumptions Behind the Linear Regression Model”. In: *SSRN Electronic Journal* (2010). ISSN: 1556-5068. DOI: [10.2139/ssrn.1584516](https://doi.org/10.2139/ssrn.1584516).
- [41] Md Kamrul Hasan et al. *Missing value imputation affects the performance of machine learning: A review and analysis of the literature (2010–2021)*. Jan. 2021. DOI: [10.1016/j.imu.2021.100799](https://doi.org/10.1016/j.imu.2021.100799).

- [42] Christina Mack, Zhaohui Su, and Daniel Westreich. *Managing Missing Data in Patient Registries: Addendum to Registries for Evaluating Patient Outcomes: A User's Guide, Third Edition*. Feb. 2018.
- [43] Minqian Yan et al. "Comparison of different approaches in handling missing data in longitudinal multiple-item patient-reported outcomes: a simulation study". In: *Health and Quality of Life Outcomes* 23 (1 Dec. 2025). ISSN: 14777525. DOI: [10.1186/s12955-025-02364-0](https://doi.org/10.1186/s12955-025-02364-0).
- [44] Rachael K. Ross, Alexander Breskin, and Daniel Westreich. "When Is a Complete-Case Approach to Missing Data Valid? The Importance of Effect-Measure Modification". In: *American Journal of Epidemiology* 189 (12 Dec. 2020), pp. 1583–1589. ISSN: 14766256. DOI: [10.1093/aje/kwaa124](https://doi.org/10.1093/aje/kwaa124).
- [45] Hyun Kang. *The prevention and handling of the missing data*. May 2013. DOI: [10.4097/kjae.2013.64.5.402](https://doi.org/10.4097/kjae.2013.64.5.402).
- [46] Jiaxu Peng, Jungpil Hahn, and Ke Wei Huang. "Handling Missing Values in Information Systems Research: A Review of Methods and Assumptions". In: *Information Systems Research* 34 (1 Mar. 2023), pp. 5–26. ISSN: 1526-5536. DOI: [10.1287/isre.2022.1104](https://doi.org/10.1287/isre.2022.1104).
- [47] J. A C Sterne et al. "Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls". In: *BMJ* 338 (jun29 1 Sept. 2009), b2393–b2393. ISSN: 0959-8138. DOI: [10.1136/bmj.b2393](https://doi.org/10.1136/bmj.b2393).
- [48] Joseph R. Dettori, Daniel C. Norvell, and Jens R. Chapman. "The Sin of Missing Data: Is All Forgiven by Way of Imputation?" In: *Global Spine Journal* 8 (8 Dec. 2018), pp. 892–894. ISSN: 21925690. DOI: [10.1177/2192568218811922](https://doi.org/10.1177/2192568218811922).
- [49] Stef van Buuren. *Flexible imputation of missing data*. CRC Press, 2019. ISBN: 0429960344.
- [50] Ian R White, Patrick Royston, and Angela M Wood. "Tutorial in Biostatistics Multiple imputation using chained equations: Issues and guidance for practice". In: (2010). DOI: [10.1002/sim.4067](https://doi.org/10.1002/sim.4067). URL: <https://onlinelibrary.wiley.com/doi/10.1002/sim.4067>.
- [51] Stef van Buuren. *Flexible Imputation of Missing Data, Second Edition*. Chapman and Hall/CRC, 2018. ISBN: 9780429492259. DOI: [10.1201/9780429492259](https://doi.org/10.1201/9780429492259).
- [52] Muhammad Arbab Arshad, Sakib Shahriar, and Khizar Anjum. "The Power Of Simplicity: Why Simple Linear Models Outperform Complex Machine Learning Techniques – Case Of Breast Cancer Diagnosis". In: (June 2023). URL: <http://arxiv.org/abs/2306.02449>.

- [53] Michael A Poole and Patrick N O’farrell. *The assumptions of the linear regression model*. Tech. rep.
- [54] J Scott Long and Laurie H Ervin. *Using Heteroscedasticity Consistent Standard Errors in the Linear Regression Model*. Tech. rep. URL: www.indiana.edu/~jslsoc/hccm.htm.
- [55] Malcolm Faddy, Nicholas Graves, and Anthony Pettitt. “Modeling length of stay in hospital and other right skewed data: Comparison of phase-type, gamma and log-normal distributions”. In: *Value in Health* 12 (2 2009), pp. 309–314. ISSN: 15244733. DOI: [10.1111/j.1524-4733.2008.00421.x](https://doi.org/10.1111/j.1524-4733.2008.00421.x).
- [56] Nuno Bastos et al. “Multiplicative Mixed-Effects Modelling of Dengue Incidence: An Analysis of the 2019 Outbreak in the Dominican Republic”. In: (2023). DOI: [10.3390/axioms12020150](https://doi.org/10.3390/axioms12020150). URL: <https://doi.org/10.3390/axioms12020150>.
- [57] Muhammad Ibrahim. “Evolution of Random Forest from Decision Tree and Bagging: A Bias-Variance Perspective”. In: *Dhaka University Journal of Applied Science and Engineering* 7 (1 Feb. 2023), pp. 66–71. ISSN: 2218-7413. DOI: [10.3329/dujase.v7i1.62888](https://doi.org/10.3329/dujase.v7i1.62888).
- [58] Jitendra Singh Kushwah et al. “Comparative study of regressor and classifier with decision tree using modern tools”. In: *Materials Today: Proceedings* 56 (Jan. 2022), pp. 3571–3576. ISSN: 2214-7853. DOI: [10.1016/j.matpr.2021.11.635](https://doi.org/10.1016/j.matpr.2021.11.635).
- [59] Hemant Ishwaran. “The Effect of Splitting on Random Forests”. In: *Machine Learning* 99 (1 Apr. 2015), pp. 75–118. ISSN: 1573-0565. DOI: [10.1007/s10994-014-5451-2](https://doi.org/10.1007/s10994-014-5451-2).
- [60] John Kitchin and Siddhant Lambor. *split random forest*. 2020. URL: <https://kitchingroup.cheme.cmu.edu/s20-06681/09-random-forests/09-random-forests.html?>
- [61] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Series in Statistics.
- [62] Hesham A. Sakr et al. “Detecting DDoS threats in IoT-driven 6G-energy hubs networks using machine learning algorithms”. In: *Discover Applied Sciences* 7 (9 Sept. 2025). ISSN: 30049261. DOI: [10.1007/s42452-025-06716-9](https://doi.org/10.1007/s42452-025-06716-9).
- [63] Gérard Biau and Erwan Scornet. “A Random Forest Guided Tour”. In: (Nov. 2015). URL: <http://arxiv.org/abs/1511.05741>.
- [64] Leo Breiman. “Random Forests”. In: (Jan. 2001).

- [65] Xi Chen and Hemant Ishwaran. *Random forests for genomic data analysis*. June 2012. DOI: [10.1016/j.ygeno.2012.04.003](https://doi.org/10.1016/j.ygeno.2012.04.003).
- [66] Gaoxia Jiang and Wenjian Wang. “Error estimation based on variance analysis of k-fold cross-validation”. In: *Pattern Recognition* 69 (Sept. 2017), pp. 94–106. ISSN: 0031-3203. DOI: [10.1016/j.patcog.2017.03.025](https://doi.org/10.1016/j.patcog.2017.03.025).
- [67] Juan D. Rodríguez, Aritz Pérez, and Jose A. Lozano. “A general framework for the statistical analysis of the sources of variance for classification error estimators”. In: *Pattern Recognition* 46 (3 Mar. 2013), pp. 855–864. ISSN: 00313203. DOI: [10.1016/j.patcog.2012.09.007](https://doi.org/10.1016/j.patcog.2012.09.007).
- [68] Sudhir Varma and Richard Simon. “Bias in error estimation when using cross-validation for model selection”. In: *BMC Bioinformatics* 7 (Feb. 2006). ISSN: 14712105. DOI: [10.1186/1471-2105-7-91](https://doi.org/10.1186/1471-2105-7-91).
- [69] Ariel Kleiner et al. *A scalable bootstrap for massive data*. Tech. rep. 2014, pp. 795–816. URL: <https://academic.oup.com/jrsssb/article/76/4/795/7076037>.
- [70] Christopher S. Withers and Saralees Nadarajah. “Analytic Bias Reduction for k-Sample Functionals”. In: (Mar. 2009). URL: <http://arxiv.org/abs/0903.2889>.
- [71] S. E. Bleeker et al. “External validation is necessary in prediction research: A clinical example”. In: *Journal of Clinical Epidemiology* 56 (9 Sept. 2003), pp. 826–832. ISSN: 08954356. DOI: [10.1016/S0895-4356\(03\)00207-5](https://doi.org/10.1016/S0895-4356(03)00207-5).
- [72] Timothy O. Hodson. *Root-mean-square error (RMSE) or mean absolute error (MAE): when to use them or not*. July 2022. DOI: [10.5194/gmd-15-5481-2022](https://doi.org/10.5194/gmd-15-5481-2022).
- [73] Min S Kim et al. *Comparative R-Squared*. Tech. rep. 2026.
- [74] *On Leakage in Machine Learning Pipelines*. Tech. rep. URL: <https://hunch.net/?p=22>.
- [75] Lin Song, Peter Langfelder, and Steve Horvath. *Random generalized linear model: a highly accurate and interpretable ensemble predictor*. Tech. rep. 2013, p. 5. URL: <http://www.biomedcentral.com/1471-2105/14/5>.
- [76] Robin Genuer et al. *Variable selection using Random Forests*. Tech. rep. 2010, pp. 2225–2236. URL: <https://hal.science/hal-00755489v1>.
- [77] Miguel Marino et al. “Missing data in primary care research: importance, implications and approaches”. In: *Family Practice* 38 (2 Apr. 2021), pp. 199–202. ISSN: 14602229. DOI: [10.1093/fampra/cmaa134](https://doi.org/10.1093/fampra/cmaa134).

- [78] Sophia M. Simon, Paul Glaum, and Fernanda S. Valdovinos. “Interpreting random forest analysis of ecological models to move from prediction to explanation”. In: *Scientific Reports* 13 (1 Dec. 2023). ISSN: 20452322. DOI: [10.1038/s41598-023-30313-8](https://doi.org/10.1038/s41598-023-30313-8).
- [79] J. W Osborne. “Notes on the use of data transformations”. In: *Practical Assessment, Research & Evaluation* 8 (6 2002). URL: <https://files.eric.ed.gov/fulltext/ED470205.pdf>.
- [80] Kerry Dwan et al. “Systematic Review of the Empirical Evidence of Study Publication Bias and Outcome Reporting Bias”. In: (). DOI: [10.1371/journal.pone.0003081](https://doi.org/10.1371/journal.pone.0003081). URL: www.plosone.org.
- [81] Nitesh V Chawla et al. *SMOTE: Synthetic Minority Over-sampling Technique*. Tech. rep. 2002, pp. 321–357.
- [82] Seba Susan and Amitesh Kumar. “The balancing trick: Optimized sampling of imbalanced datasets-A brief survey of the recent State of the Art”. In: (2020). DOI: [10.1002/eng2.12298](https://doi.org/10.1002/eng2.12298). URL: <https://doi.org/10.1002/eng2.12298>.